

Student Name: Maths Teacher

SYDNEY TECHNICAL HIGH SCHOOL



HSC ASSESSMENT TASK 1 DECEMBER 2004

MATHEMATICS

Time allowed: 70 minutes

Instructions

- * Write your details at the top of this page.
- * Attempt all questions. All questions are worth equal marks.
- * Answers are to be written on the paper provided.
- * Do **not** divide your pages into two columns of working.
- * You may write on the front and back of each page. Ask for more paper if required.
- * Marks may not be awarded for careless or badly arranged working.
- * Indicated marks are a guide and may be changed slightly if necessary.

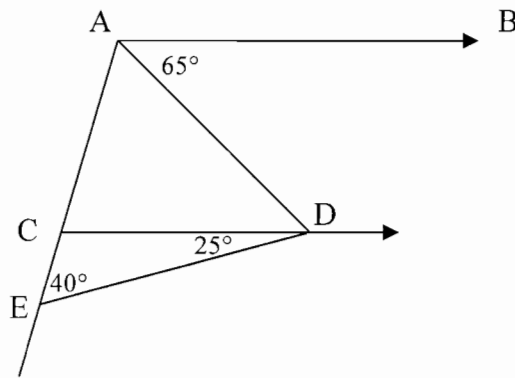
Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
/7	/7	/7	/7	/7	/7	/7	/7	/56

Question 1

- 3 a) A parabola has its focus at $(0, 2)$ and its directrix is the line $y = -4$. Find
- i) the coordinates of the vertex.
 - ii) the equation of the parabola.
- 4 b) The roots of the equation $x^2 - 8x + 10 = 0$ are α and β . Find
- i) $\alpha + \beta$
 - ii) $\alpha\beta$
 - iii) $\alpha^2 + \beta^2$

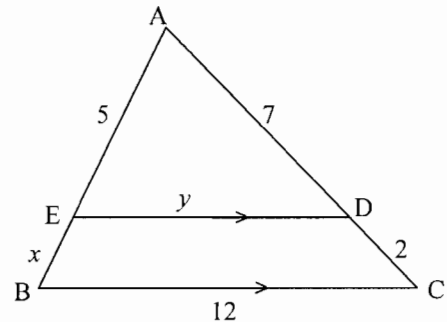
Question 2 (Begin a new page)

- 4 a) A parabola has equation $y = x^2 - 4x - 21$. Find
- the equation of the axis of symmetry.
 - the coordinates of the vertex.
 - the x intercept/s.
 - the values of x for which $x^2 - 4x - 21 < 0$.
- 3 b) Copy this diagram onto your page. $AB \parallel CD$. Prove that triangle ACD is isosceles.



Question 3 (Begin a new page)

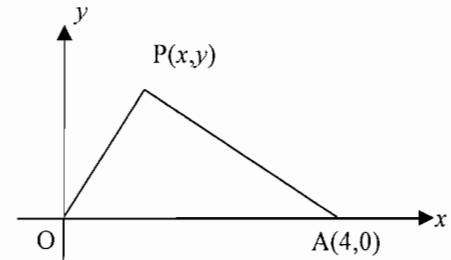
- 4 a) In this part **no** formal proofs are required but you must give full reasons for the statements you make.
- Find the value of x giving reasons.
 - Find the value of y giving reasons.



- 3 b) A parabola has equation $x^2 - 4y - 4x + 16 = 0$.
- Write the equation in the form $(x - h)^2 = 4a(y - k)$.
 - Find the coordinates of the vertex.
 - Find the coordinates of the focus.

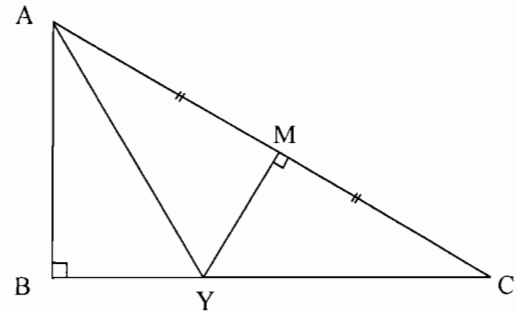
Question 4 (Begin a new page)

- 2 a) Solve for x : $x^4 - 6x^2 + 8 = 0$.
- 5 b) For the diagram at the right:
- Write expressions for the gradients of AP and OP.
 - If $\angle OPA$ is always 90° , show that the equation of the locus of P represents a circle.
 - State the centre and radius of the circle.



Question 5 (Begin a new page)

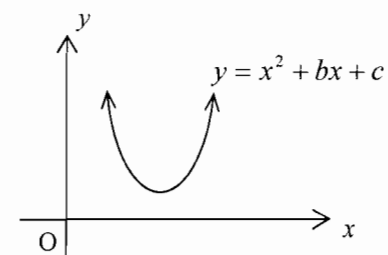
- 3 a) i) Sketch the graph of $(y - 2)^2 = 8(x + 1)$ showing clearly the coordinates of the vertex..
- ii) Draw and label the directrix and write its equation on the sketch.
- 4 b) In $\triangle ABC$, $\angle B = 90^\circ$. YM is the perpendicular bisector of AC. Copy the diagram onto your page.
- Prove that $\triangle AYM \equiv \triangle CYM$.
 - Suppose now that AY bisects $\angle BAC$. Find the size of $\angle YCM$ (no reasons needed).



Question 6 (Begin a new page)

- 2 a) i) Write a quadratic equation with roots α and β if $\alpha + \beta = -2$ and $\alpha\beta = 6$.
- ii) Write a quadratic equation with roots k and $2k$.
- 3 b) By making a suitable substitution find all real values of x which satisfy the equation $(x^2 - 1)^2 - 3(x^2 - 1) = 0$.

- 2 c) i) State the condition, in terms of b & c , for the graph of $y = x^2 + bx + c$ to be entirely above the x axis.
- ii) What two word description is used for quadratics of this type?

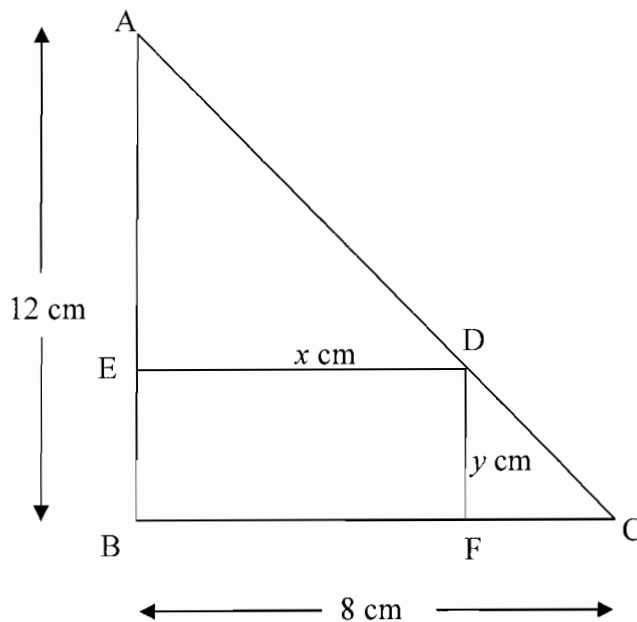


Question 7 (Begin a new page)

- 3 a) Find the value/s of k for which the equation $x^2 + 4x + k = 0$ has roots which are real and distinct (unequal).
- 4 b) Find possible values of m so that the line $y = mx - 9$ will be a tangent to the curve $y = x^2 - 2x$.

Question 8 (Begin a new page)

EDFB is a rectangle with sides x and y inscribed in the triangle ABC . Side AB is 12 cm in length and side BC is 8 cm in length.



- 1 a) Which test would be used to show that $\triangle ABC \sim \triangle DFC$?
- 2 b) Show that $\frac{8-x}{y} = \frac{8}{12}$ (give a reason).
- 2 c) Show that the area of the rectangle is $12x - \frac{3}{2}x^2$.
- 2 d) Use the theory of quadratic functions (not calculus) to find the value of x which makes the rectangle area a maximum and find this maximum area.

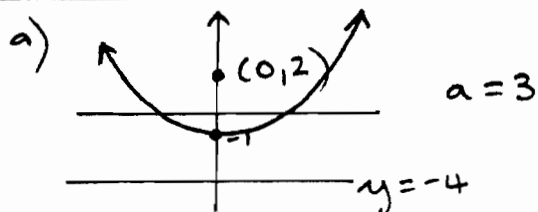
End of Questions.

Place this question paper on top of your answers to hand in.

MARKING SCHEME.

STHS 2 UNIT HSC TASK 1 DEC 2004

Question 1



- i) Vertex $(0, -1)$
 ii) $x^2 = 12(y + 1)$

a) i) ① mark

ii) ① for correct form: $x^2 = 4ay$
 ① for correct equation.

b) i) $\alpha + \beta = \frac{-b}{a} = \underline{8}$

ii) $\alpha\beta = \frac{c}{a} = \underline{10}$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 8^2 - 2 \times 10$
 $= \underline{44}$

b) i) ① mark

ii) ① mark

(iii) ② for correct answer
 OR
 ① for correct formula

allow errors
 carried
 through

Question 2

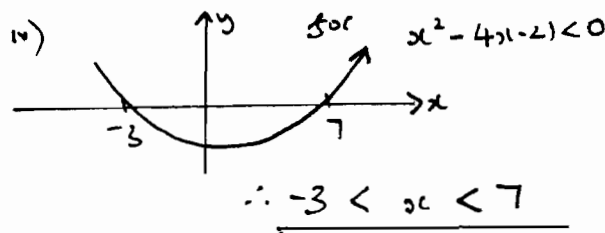
a) $y = x^2 - 4x - 2$

i) axis of sym $x = \frac{4}{2}$
 $\therefore x = \underline{2}$

ii) Vertex $(2, -25)$

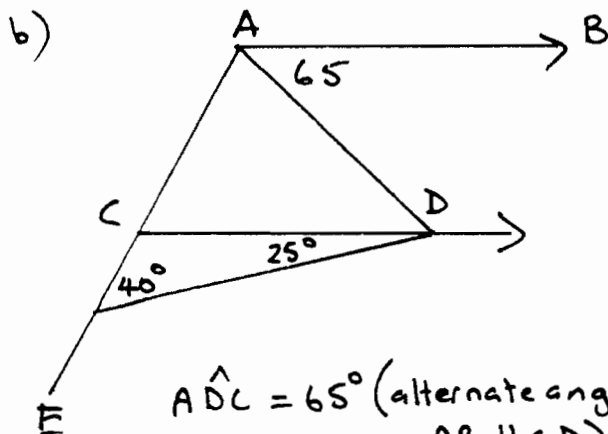
iii) $y = (x - 7)(x + 3)$

$\therefore x$ intercepts $x = 7$ and -3



① mark each part

Allow errors carried forward
 from i) to ii) and from
 iii) to iv)



$$\hat{ADC} = 65^\circ \text{ (alternate angles, } AB \parallel CD) \text{ — ① all correct}$$

$$\hat{ACD} = 65^\circ \text{ (exterior angle of triangle)} \text{ — ① all correct}$$

$\therefore \triangle ACD$ is isosceles

(base angles equal)

— ① Correct reason — allow (2 angles equal etc)

Question 3

a) i) $\frac{5}{x} = \frac{7}{2}$ (ratio of intercepts, parallel lines)

$$7x = 10$$

$$x = \frac{10}{7}$$

ii) $\frac{y}{12} = \frac{7}{9}$ (similar triangles corresponding sides in proportion)

$$9y = 84$$

$$y = 9\frac{1}{3}$$

a) i) and ii)

① for correct answer

— ① for reason which fits the equation — Reason must clearly identify the theorem used for the written equation.

— DITTO

b) i) $x^2 - 4x = 4y - 16$

$$x^2 - 4x + 4 = 4y - 16 + 4$$

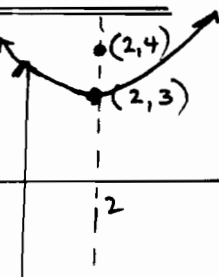
$$(x-2)^2 = 4y - 12$$

$$(x-2)^2 = 4(y-3)$$

ii) Vertex (2, 3)

iii) $a = 1$

Focus (2, 4)



b) ① for each part.

Allow errors carried from (i) to (ii) and (iii).

Question 4

a) Let $u = x^2$

$$u^2 - 6u + 8 = 0$$

$$(u-4)(u-2) = 0$$

$$u = 4$$

$$u = 2$$

$$x^2 = 4$$

$$x^2 = 2$$

$$x = \pm 2$$

$$x = \pm \sqrt{2}$$

① to here

① for this operation as long as 1 value of u has been correctly solved for x .

b) i) $m_{OP} = \frac{y}{x}$

$m_{PA} = \frac{y}{x-4}$

i) ① both must be correct

ii) $\frac{y}{x} \cdot \frac{y}{x-4} = -1$ ①

ii) → for correct statement allow E.C.F. from (i).

$$y^2 = -x(x-4)$$

$$y^2 = -x^2 + 4x$$

$$4 - 4x + x^2 + y^2 = 4$$

ie. $(x-2)^2 + y^2 = 4$ which is a circle

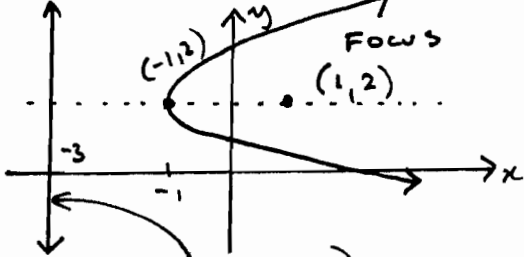
① for rearranging into standard circle - even if incorrect eqn.

(iii) centre (2,0)
radius 2

① each. Allow if correctly deduced from an incorrect equation in (ii).

Question 5

a) i) vertex $(-1, 2)$ $a = 2$



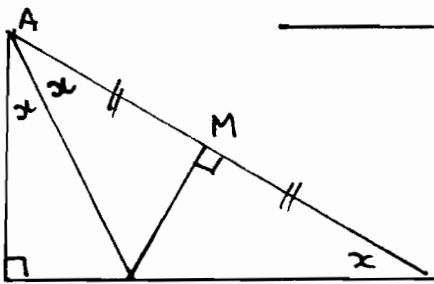
① for correct shape

① for correct vertex

ii) $x = -3$ (directrix)

① for correct directrix & eqn.

b)



QUESTION 5 (cont)

i) In Δ 's AYM , CYM

$AM = MC$ (M is midpt of AC)

$\hat{A}MY = \hat{C}MY$ (supplementary angles
and $YM \perp AC \therefore$
both 90°)

MY is common

$\therefore \underline{\underline{\Delta AYM \equiv \Delta CYM \text{ (SAS)}}$

ii) Let $\hat{B}AY = x \therefore \hat{Y}AM = x$

(AY bisects $\hat{B}AC$)

and $\hat{M}CY = x$ (corresp. angles
in congruent triangles)

$$\therefore 3x = 90$$

$$\underline{\underline{\hat{Y}CM = x = 30^\circ}}$$

③ for correct proof

ie ① for each correct line

- ignore conclusion.

① for correct value (30°) ($\frac{\pi}{6}$)
no working needed. Allow if
degree sign missing.

Question 6

a) i) $x^2 + 2x + 6 = 0$

①

ii) $(x-k)(x-2k) = 0$

① No need to expand.

b) $u = x^2 - 1$

$$u^2 - 3u = 0$$

① Simplified equation

$$u(u - 3) = 0$$

$$u = 0$$

$$u = 3$$

① Correct solutions

$$x^2 - 1 = 0$$

$$x^2 - 1 = 3$$

$$x^2 = 1$$

$$x^2 = 4$$

$$x = \pm 1$$

$$x = \pm 2$$

① 1 value of u correctly
followed through.

c)

i) $\therefore (\Delta < 0) \quad \underline{\underline{b^2 - 4c < 0}}$

① must be in terms of b & c .

ii) positive definite

① Allow mis spelling

Question 7

a) roots real and unequal

$$\therefore \Delta > 0$$

$$4^2 - 4 \times 1 \times k > 0$$

$$16 - 4k > 0$$

$$16 > 4k$$

$$4 > k$$

$$\therefore k < 4$$

① for rule

① for correctly finding Δ
(even if inequality is wrong)

① Correct solution of their inequality.

b) tangent if one solution
when solved simultaneously

$$\begin{cases} y = mx - 9 \\ y = x^2 - 2x \end{cases}$$

$$mx - 9 = x^2 - 2x$$

$$0 = x^2 - 2x - mx + 9$$

$$0 = x^2 - x(2+m) + 9$$

$\therefore \Delta = 0$ for this quadratic

$$(2+m)^2 - 4 \times 1 \times 9 = 0$$

$$4 + 4m + m^2 - 36 = 0$$

$$m^2 + 4m - 32 = 0$$

$$(m+8)(m-4) = 0$$

If $m = -8$ $m = 4$ line is
tangent to curve.

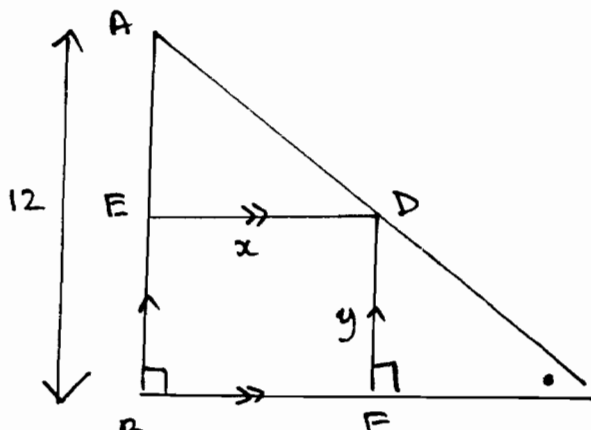
① for either of these

① initial substitution
Correct.

① Correct inequality (even if
from an incorrect quadratic eqn.)

① Correct solution of
their inequality.

Question 8

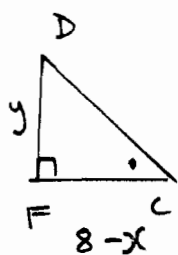
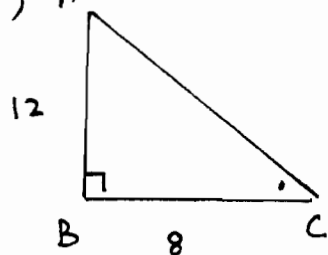


QUESTION 8 (cont)

a) $\triangle ABC \parallel \triangle DFC$ (equiangular)

①

b)



① $FC = 8-x$ must be stated or indicated (eg. on a diagram) etc.

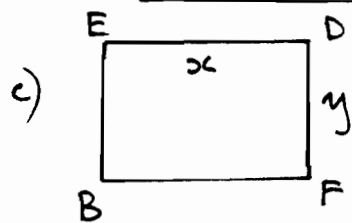
① for suitable reason.

No marks for writing the given statement.

$$\frac{8}{8-y} = \frac{12}{y} \quad (\text{corr sides similar triangles})$$

$$8y = 12(8-x)$$

$$\frac{8}{12} = \frac{8-x}{y}$$



$$8y = 12(8-x)$$

$$\therefore y = \frac{12(8-x)}{8} \quad \text{①}$$

$$y = \frac{3(8-x)}{2}$$

Both must be correct. No E.C.F. allowed.

$$\therefore \text{area rectangle} = x \times 3 \frac{(8-x)}{2} \quad \text{①}$$

$$= 12x - \frac{3x^2}{2}$$

d) Let

$$A = -\frac{3x^2}{2} + 12x$$

→ Alternatively

$$A = -\frac{3}{2}(x^2 - 8x + 16) + 24$$

max of 24 when x = 4

axis of sym. $x = \frac{-12}{2 \times -\frac{3}{2}}$

$$x = 4 \quad \text{①}$$

$\therefore$ max area is

$$= 12 \times 4 - \frac{3}{2}(4)^2$$

$$= \underline{\underline{24 \text{ cm}^2}} \quad \text{① for correct value (24)}$$